

HS 650: Statistical Methods

Course Code	HS 650
Course Name	Statistical Methods
L-T-P-C	3-0-0-3
Prerequisites	A prior course in probability, statistics and random processes; or, discretion of the instructors
Intended for	Postgraduate and B.Tech.
Distribution	HSS Course Elective
Approval	10th Senate (Jun 2016)

Course Contents

- **Representation of Data and Descriptive Statistics** Raw data and frequency data- tabular and diagrammatic representation; concept of moments; measures of central tendency, dispersion, skewness, and Kurtosis; quartile and percentile their use in the measurement of inequality, Gini Coefficient and Lorenz curve; Bivariate frequency distribution, correlation coefficients- Pearson and Spearman coefficients. [6 Lectures]
- **Probability and Random Variables** Basic concepts in set theory as applied in probability; concept of probability- classical, frequency based, axiomatic approach, Bayesian probability; conditional probability, Bayes theorem, statistical independence of events; random variables discrete and continuous, probability distribution functions, cumulative distribution functions, Expectation and Variance of a random variable, joint distribution of two random variables and their correlation, law of large number. [11 Lectures]
- **Random Sampling and Parametric Statistical Inference** Concepts of population and sample, parameter and statistic, random sampling and sampling distribution, Central Limit Theorem; Expectation and Standard Error of sample mean and sample proportion; concepts of theoretical distribution: Normal distributions and four fundamental distributions derived from Normal distribution Standard Normal, Chi-square, t and F distribution; estimation and testing of hypothesis point estimation and interval estimation of parameters, Maximum Likelihood Estimator, hypothesis testing, and calculation of effect size. [9 Lectures]
- **Non-parametric Statistical Inference** Need for non-parametric tests, estimation of location and dispersion, tolerance interval; one sample and two sample non-parametric tests for location and dispersion (involving independent and related samples); non-parametric measures and tests of association. [6 Lectures]
- **Designs of Experiment** Experimental design strategies; Blocking and Randomization; Factorial design of experiments. [6 Lectures]
- **Regression Analysis and Analysis of Variance** Gauss Markov theorem and Ordinary Linear Least Square regression; interpreting regression coefficients, concepts of residual, fitted value and goodness of fit, test of significance; multiple regression analysis; two-way independent ANOVA and two-way Mixed ANOVA [8 Lectures]

Text Books

1. Field, A. P., Miles, J., & Field, Z., **Discovering statistics using R**, Sage, 2012.
2. Field, A. P., **Discovering statistics using IBM SPSS Statistics**, Sage, 2013.

References

1. Agresti, A., & Finlay, B., **Statistical Methods for the Social Sciences**, 1997.
2. Arnold, J. C., & Milton, J. S., **Introduction to probability and statistics**, 2003.
3. Gibbons, J. D., Chakraborti, S., **Nonparametric Statistical Inference**, 5th Edition, Marcel Dekker, 2003.
4. Johnston, J., DiNardo, J., **Econometric Methods**, 4th Edition. McGraw-Hill, 1996
5. Montgomery, D. C., G.C. Runger, **Applied Statistics and Probability for Engineers**, 5th Edition, Wiley-India, 2011.
6. Montgomery, D. C., **Design and Analysis of Experiments**, 8th Edition. John Wiley & Sons, 2012.
7. Ross, S. M., **Introduction to probability and statistics for engineers and scientists**, Academic Press, 2014.
8. Rohatgi, V. K. & Saleh, A. K. E., **An Introduction to Probability and Statistics**, 2015.